PS-4 run report: candidate reduction by Frobenius-orbit representatives and the 2-torsion symmetry

PDP oracle speedup program, `research/pdp_oracle_speedup_20261011/ps4_symmetry/` — run started 2026-10-11 23:46 UTC, measurements finished 2026-10-25; frozen protocol `protocols/PS-4.md`; class (as frozen): engineering; stage measurement

1. What ran

`ps4_symmetry/run_ps4.sh` (stages build, checks, panel, run, torsion, control, summary, figures, pdf) with the standalone example `examples/pdp_ps4`, the panel `examples/hamming_nb_sat_panel` for its exact oracle, and the PS-2 group-law kernel on the PS-0 corpus (library counters, `--features pdp-opcount`) as the subspace-leaf control. Every build and run went through `tools/isolated_bench.py busy --wait`. Conductor: `conductor check` was clear; the checkout has no `.conductor/WORKFLOW.md` or task card.

Workload. Fourteen settings, the Hamming-weight base `wt <= w` with `w` in {2, 3} at the type-II degrees `n` in {11, 23, 29, 33, 35, 39, 41} of the curve `K_0: y^2 + xy = x^3 + 1`; per setting the panel's 8 targets (6 uniform, 2 planted sums of three base points; seed 20261011; `m = 3`): 112 targets. Raw records: `ps4_symmetry/raw/*.jsonl`; tables: `raw/summary.md`.

Hosts. The checks, the panel runs and the unreduced/reduced runs up to `(n, w) = (39, 3)` and at `(41, 2)` ran on an Intel Xeon 2.80 GHz KVM guest (kernel v114, 4 vCPU, AVX-512, `pclmulqdq`); a container restart moved the rest (run `(41, 3)`, torsion, control) to an Intel Xeon 2.10 GHz guest (kernel v139); a second restart interrupted the torsion stage at `(33, 3)`, whose partial records are kept in `raw/attempts/`; a third container (same CPU class, kernel v153) repeated the run at `(35, 3)` with the panel's records and built the report. Every count below is deterministic and host independent; wall figures are labelled and never pooled across the two hosts.

2. Observations

2.1. Candidates and counted operations, reduced against unreduced

Full-scan mode, sums over the 8 targets of a setting. The unreduced loops take every base abscissa as first summand (`ordered`: every abscissa as second summand, `m^2` pairs; `unordered`: from the first on, `m(m+1)/2` pairs, the PS-0/PS-2 loop); the reduced loop takes one representative per Frobenius orbit as first summand and every abscissa as second (`reps * m` pairs). `m` = abscissae of weight `<= w` that lift. `M` = counted field multiplications (`mul` + `batch_mul`, inversion products included).

n	w	m	reps	short orbits	pairs ordered	pairs unordered	pairs reduced	factor ordered	factor unordered	M factor unordered	M factor ordered	op/pair unordered	op/pair ordered
11	2	23	3	(0x0, 1)	529	276	69	7.667	4.000	4.226	7.645	1.056	0.997
23	2	162	8	(0x0, 1)	26244	13203	1296	20.250	10.188	10.322	20.250	1.013	1.000
29	2	175	7	(0x0, 1)	30625	15400	1225	25.000	12.571	12.726	25.000	1.012	1.000
33	2	232	8	(0x0, 1)	53824	27028	1856	29.000	14.562	14.680	29.000	1.008	1.000
35	2	211	7	(0x0, 1)	44521	22366	1477	30.143	15.143	15.298	30.143	1.010	1.000
39	2	430	12	(0x0, 1)	184900	92665	5160	35.833	17.958	18.062	35.833	1.006	1.000
41	2	288	8	(0x0, 1)	82944	41616	2304	36.000	18.062	18.198	36.000	1.008	1.000
11	3	100	10	(0x0, 1)	10000	5050	1000	10.000	5.050	5.141	10.042	1.018	1.004
23	3	1197	53	(0x0, 1)	1432809	717003	63441	22.585	11.302	11.323	22.585	1.002	1.000
29	3	2031	71	(0x0, 1)	4124961	2063496	144201	28.606	14.310	14.325	28.606	1.001	1.000
33	3	3246	100	(0x0, 1) (0x142, 11)	10536516	5269881	324600	32.460	16.235	16.244	32.460	1.001	1.000
35	3	3151	91	(0x0, 1)	9928801	4965976	286741	34.626	17.319	17.331	34.626	1.001	1.000
39	3	4993	129	(0x0, 1)	24930049	12467521	644097	38.705	19.357	19.366	38.705	1.001	1.000
41	3	5372	132	(0x0, 1)	28858384	14431878	709104	40.697	20.352	20.361	40.697	1.000	1.000

`short orbits` lists the orbits smaller than `n` (the point `T`, `x = 0`, with size 1 in every setting; at `(33, 3)` also a period-11 orbit of size 11); orbit sizes add up to `m` in all 14 settings. `op/pair` = (M factor) / (pair factor): the fraction of the candidate factor that the counted multiplications keep (the frozen criterion asks for at least 0.8).

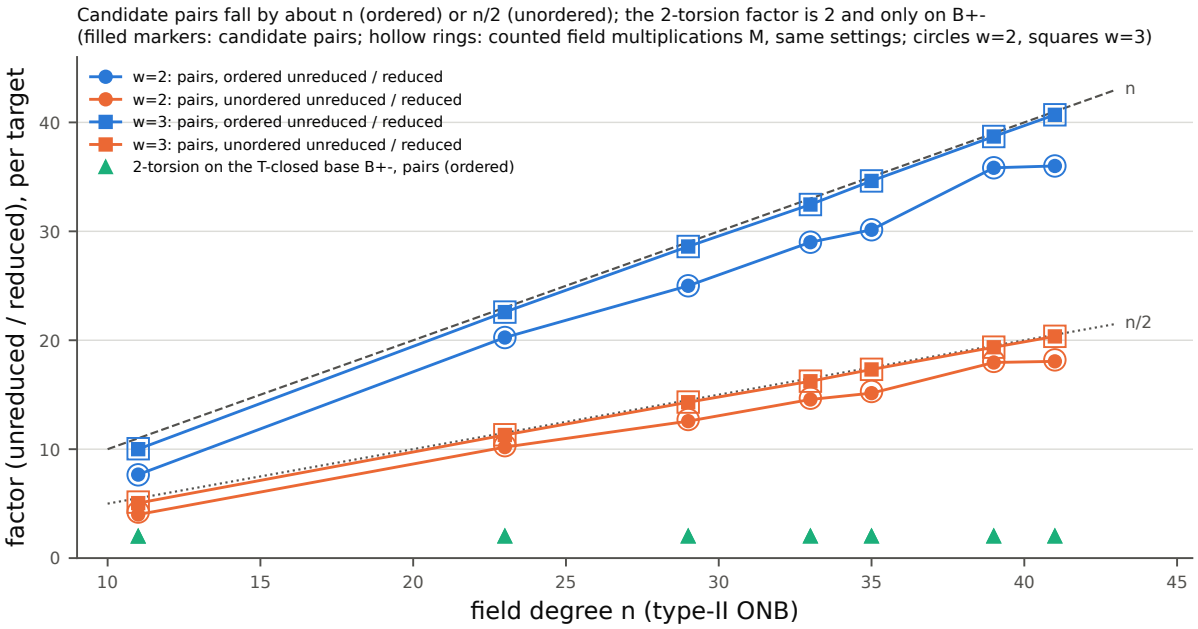


Figure 1: Candidate-pair factor, unreduced over reduced, per setting (source `raw/plot_data.json`, derived from `raw/runs.jsonl`, `raw/torsion.jsonl`; 8 targets per setting, 2 at the three largest `w = 3` torsion settings). Filled markers: pairs. Hollow rings: counted multiplications `M`. Triangles: the 2-torsion reduction, measured on the T-closed base `B+-`, not on the weight base.

Counted operations per pair are the same in every loop and setting: `mul` 4.00 to 5.35, `sqr` 4.0 to 4.7 (a coordinate permutation here), `batch_mul` 6.00, `inv` 0.0006 to 0.25, `add` 16, `popcnt` 4, `M` 10.0 to 11.3 (`raw/summary.md`, Table 2b).

2.2. Verdicts and solution sets

n	w	targets	decomposable	panel xR	panel verdict	panel ref_count	U-unord = oracle	U-ord = oracle	R subset	R = predicted	kept solutions
11	2	8	5	8/8	8/8	8/8	8/8	8/8	8/8	8/8	21/66 = 0.318 (independence: 0.342)
11	3	8	8	8/8	8/8	8/8	8/8	8/8	8/8	8/8	1240/4797 = 0.258 (independence: 0.271)
23	2	8	6	8/8	8/8	8/8	8/8	8/8	8/8	8/8	3/19 = 0.158 (independence: 0.141)
23	3	8	8	8/8	8/8	8/8	8/8	8/8	8/8	8/8	250/2034 = 0.123 (independence: 0.127)
29	2	8	2	8/8	8/8	8/8	8/8	8/8	8/8	8/8	0/2 = 0.000 (independence: 0.115)
29	3	8	8	8/8	8/8	8/8	8/8	8/8	8/8	8/8	21/165 = 0.127 (independence: 0.101)
33	2	8	2	8/8	8/8	8/8	8/8	8/8	8/8	8/8	0/2 = 0.000 (independence: 0.100)
33	3	8	8	8/8	8/8	8/8	8/8	8/8	8/8	8/8	5/76 = 0.066 (independence: 0.090)
35	2	8	2	8/8	8/8	8/8	8/8	8/8	8/8	8/8	0/2 = 0.000 (independence: 0.096)
35	3	8	6	8/8	8/8	8/8	8/8	8/8	8/8	8/8	2/15 = 0.133 (independence: 0.084)
39	2	8	2	8/8	8/8	8/8	8/8	8/8	8/8	8/8	1/2 = 0.500 (independence: 0.081)
39	3	8	5	n/a	n/a	n/a	8/8	8/8	8/8	8/8	1/9 = 0.111 (independence: 0.076)
41	2	8	2	8/8	8/8	8/8	8/8	8/8	8/8	8/8	1/2 = 0.500 (independence: 0.081)
41	3	8	2	n/a	n/a	n/a	8/8	8/8	8/8	8/8	0/2 = 0.000 (independence: 0.072)

Over the 112 targets: the oracle calls 66 decomposable and 46 not; the unreduced kernel's solution set equals the oracle's on 112/112 (unordered and ordered loops); the panel (run on 12 of the 14 settings, 96 targets) agrees with the pair-table oracle of the example on `x_R`, verdict and `ref_count` on 96/96. The reduced loop's set is contained in the oracle's on 112/112 and equals the oracle's triples that have a representative member on 112/112 (`kept solutions`: pooled distinct solutions of the reduced loop over the oracle's).

n	w	decomposable	verdict kept (target alone)	R says no	M to witness R / U	refuting	M full scan R / U	closure = U
11	2	5	8/8	0	1.014	3	0.236	8/8
11	3	8	8/8	0	1.000	0	n/a	8/8
23	2	6	5/8	3	0.308	2	0.097	8/8
23	3	8	8/8	0	1.001	0	n/a	8/8
29	2	2	6/8	2	0.188	6	0.079	8/8
29	3	8	7/8	1	1.169	0	n/a	8/8
33	2	2	6/8	2	0.134	6	0.068	8/8
33	3	8	2/8	6	0.368	0	n/a	8/8
35	2	2	6/8	2	0.142	6	0.065	8/8
35	3	6	4/8	4	0.128	2	0.058	8/8
39	2	2	7/8	1	0.102	6	0.055	8/8
39	3	5	4/8	4	0.140	3	0.052	8/8
41	2	2	7/8	1	1.142	6	0.055	8/8
41	3	2	6/8	2	0.128	6	0.049	8/8

The reduced loop on the target alone returned the unreduced verdict on 84 of 112 targets; on the 66 decomposable targets it answered no on 28. Taking all conjugates of the target (`closure = U`), the union of the reduced loop's solutions, mapped back, equals the unreduced set on 112/112; that run visits 1.007 to 1.435 times the pairs of the ordered unreduced loop on the single target (Table 5 of `raw/summary.md`). Hits per candidate third, pooled: reduced over unreduced = 1.000 at `(11, 3)` (14,505 and 2,872 hits) and 0.992 at `(23, 3)` (6,105 and 536 hits); elsewhere the hit counts are 0 to 496 and the ratio is not informative.

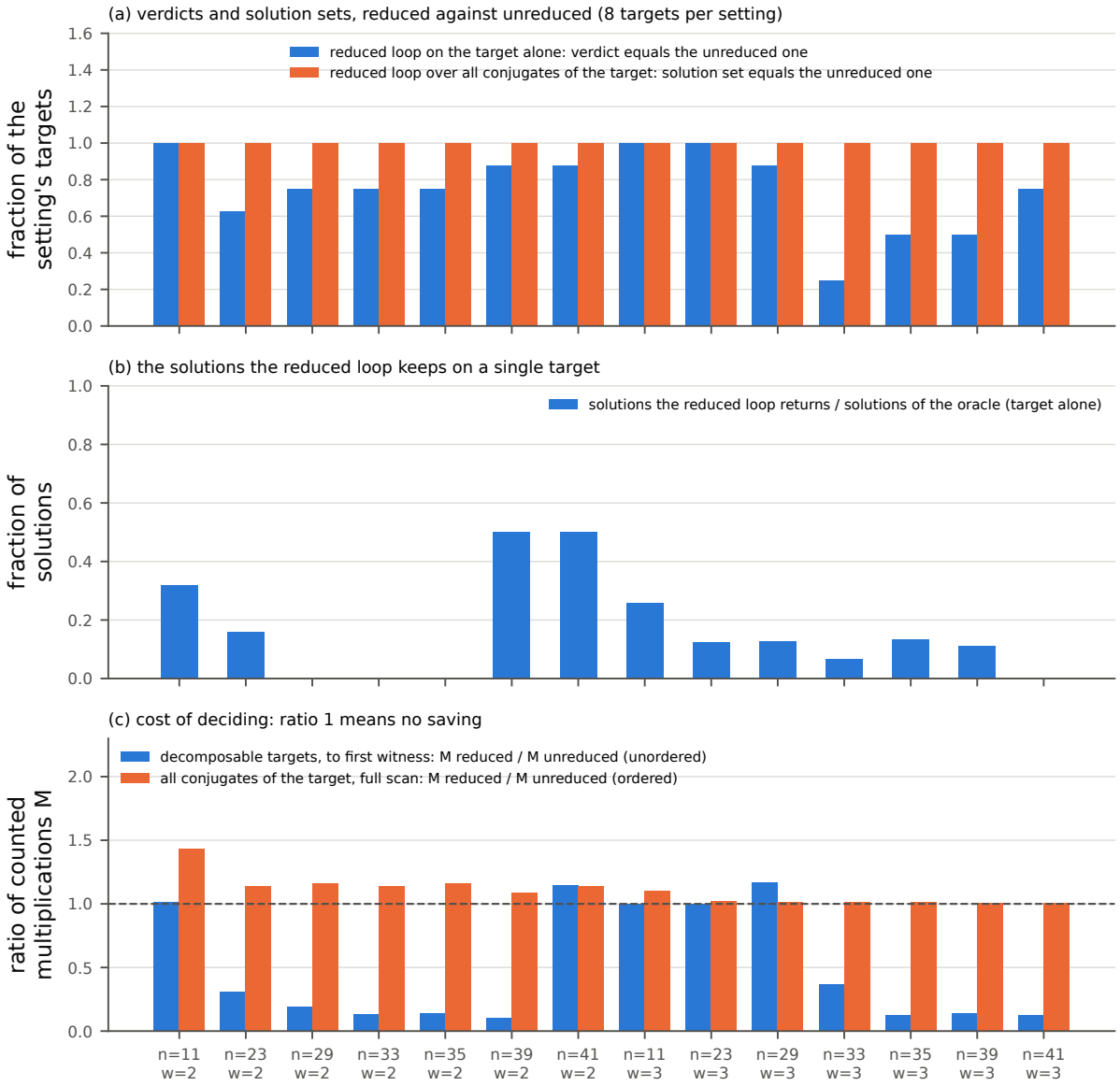


Figure 2: (a) fraction of each setting's 8 targets whose verdict the reduced loop keeps when run on the target alone, and whose solution set the reduction over all conjugates reproduces; (b) pooled fraction of the oracle's solutions the reduced loop returns on the target alone; (c) ratio of counted multiplications, reduced over unreduced, to the first witness on decomposable targets (the reduced loop's full scan counted where it found nothing), and for the all-conjugates run against the ordered unreduced loop. Source `raw/plot_data.json`.

2.3. The 2-torsion symmetry

Translation by `T = (0, 1)` is `x -> 1/x` (checked: 0 mismatches, below); in `u = 1/(x + 1)` it is `u -> u + 1`, and in the ONB `1` is the all-ones word, so it is the complement, of weight `n - wt(u)`. On the weight base, for liftable abscissae: the image of `T` is in the base for 0 of the base's abscissae in all 14 settings, in the `x` frame and in the `u` frame (columns below). The reduction was tested on the nearest `T`-closed base `B+- = {wt(u) <= w or wt(u) >= n - w}` (ordered loops, leaf a table probe; `T` alone forms a class, `T + T = 0`):

n	w	x-frame base	x-frame image in base	u-frame base	u-frame image in base	targets	m(B+-)	classes	pair factor	M factor	closure = oracle	verdict kept
11	2	22	0	44	0	8	89	45	1.9778	1.9711	8/8	8/8
23	2	161	0	161	0	8	323	162	1.9938	1.9938	8/8	8/8
29	2	174	0	232	0	8	465	233	1.9957	1.9957	8/8	8/8
33	2	231	0	363	0	8	727	364	1.9973	1.9973	8/8	8/8
35	2	210	0	280	0	8	561	281	1.9964	1.9964	8/8	8/8
39	2	429	0	390	0	8	781	391	1.9974	1.9974	8/8	8/8
41	2	287	0	328	0	8	657	329	1.9970	1.9970	8/8	8/8
11	3	99	0	132	0	8	265	133	1.9925	1.9901	8/8	8/8
23	3	1196	0	1012	0	8	2025	1013	1.9990	1.9990	8/8	8/8
29	3	2030	0	1972	0	8	3945	1973	1.9995	1.9995	8/8	8/8

n	w	x-frame base	x-frame image in base	u-frame base	u-frame image in base	targets	m(B+-)	classes	pair factor	M factor	closure = oracle	verdict kept
33	3	3245	0	2816	0	8	5633	2817	1.9996	1.9996	8/8	8/8
35	3	3150	0	3430	0	2	6861	3431	1.9997	1.9997	2/2	2/2
39	3	4992	0	4927	0	2	9855	4928	1.9998	1.9998	2/2	2/2
41	3	5371	0	5330	0	2	10661	5331	1.9998	1.9998	2/2	2/2

Here `closure` is the `T`-reduced set closed under translating two non-`T` summands by `T`, compared with the pair-table oracle on `B+-`. Not run: `T` on a Frobenius-invariant subspace containing 1 (the repository's own `u`-frame base); the combined reduction by orbits and `T` as a loop.

2.4. The leaf, weight base against the subspace control

The control is the PS-2 group-law kernel on the first 6 uniform and 2 planted targets of each PS-0 rung `l = 8..12` (`n = 24..36`, `a = b = 1`, polynomial basis; 66,048 to 15,880,224 pairs per rung): `M/pair` = 10.53, 10.24, 10.15, 10.06, 10.03 and `popcnt`/compare 4 per pair; verdicts equal the corpus on 8/8 targets of every rung. `bits` is the model of `model.rs` (a convention: schoolbook two-input gate counts, ONB product `3 n^2 - 2 n`, no sub-quadratic products); the last-but-one column puts the control at the weight base's degree by `(n / n_l)^2`.

n	w	M/pair (ONB)	bits/pair (ONB, model)	n-near l	dn	M/pair weight / control	bits weight / control	bits, control scaled to n	range over l=8..12
11	2	11.3438	4132.1	8	+13	1.078	0.312	1.484	0.154 to 0.312
11	3	10.3809	3803.4	8	+13	0.986	0.287	1.366	0.142 to 0.287
23	2	10.2699	16378.0	8	+1	0.976	1.236	1.345	0.612 to 1.236
23	3	10.0367	16018.6	8	+1	0.953	1.209	1.316	0.599 to 1.209
29	2	10.2500	25962.2	10	+1	1.010	1.375	1.472	0.970 to 1.959
29	3	10.0217	25399.4	10	+1	0.988	1.346	1.440	0.949 to 1.916
33	2	10.1631	33324.1	11	+0	1.011	1.475	1.475	1.245 to 2.514
33	3	10.0117	32839.5	11	+0	0.996	1.454	1.454	1.227 to 2.478
35	2	10.2075	37638.2	12	+1	1.017	1.407	1.488	1.407 to 2.840
35	3	10.0140	36940.3	12	+1	0.998	1.381	1.461	1.381 to 2.787
39	2	10.1160	46306.3	12	−3	1.008	1.731	1.475	1.731 to 3.494
39	3	10.0100	45830.9	12	−3	0.998	1.713	1.459	1.713 to 3.458
41	2	10.1522	51349.3	12	−5	1.012	1.919	1.480	1.919 to 3.874
41	3	10.0082	50634.6	12	−5	0.997	1.892	1.459	1.892 to 3.820

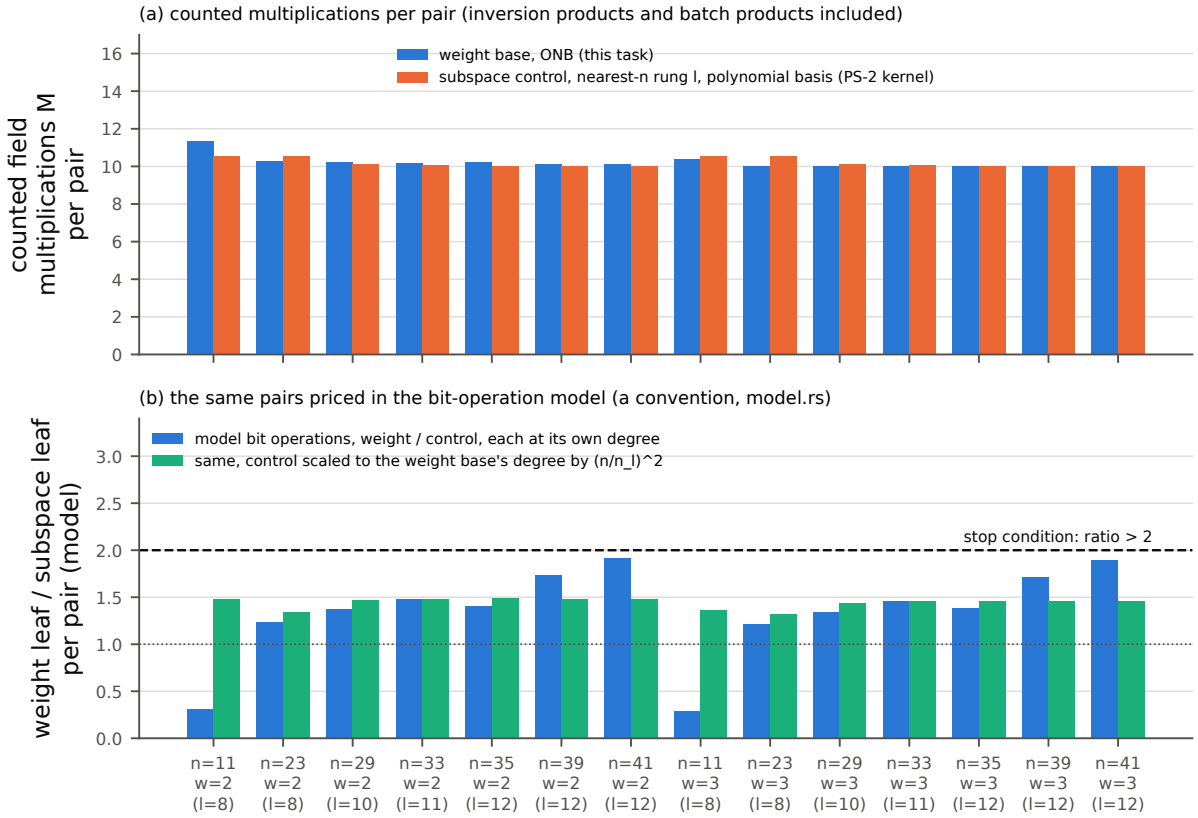


Figure 3: Leaf cost per pair, weight base in the ONB against the subspace control at the rung with `3 l` nearest `n`. (a) counted multiplications; (b) the same pairs in the bit-operation model, at each side's own degree and with the control scaled to `n`; the dashed line is the frozen stop condition (ratio above 2). Source `raw/plot_data.json`.

Wall time, a labelled note, unisolated, shared container. Weight kernel (portable software cyclic-convolution product): 480 to 1,233 ns per pair on the 2.80 GHz host and 729 and 911 ns at `(35, 3)` and `(41, 3)` on the 2.10 GHz class; control (`pclmulqdq`): 26 to 39 ns per

pair on the 2.10 GHz host. The one same-host pair, `(41, 3)` against `l = 12`: 911 against 31 ns per pair. The two kernels are different code on different instructions, so this does not compare the bases.

2.5. Algebra and field checks (`K_0`, `a = 0`, `b = 1`)

- ONB field against the library's `Type2Onb`: products exhaustive for `n <= 11` (4,194,304 pairs at `n = 11`) and 200,000 samples at each of `n = 23..41`: 0 mismatches; squarings, inversions (Itoh–Tsujii, products counted = formula) and lifts: 0 mismatches.
- Group law with `a = 0` against the library's `OnbCurve::add`: 257,556 pairs exhaustive at `n = 9`, 200,000 sampled at each of `n = 11..41`: 0 mismatches. Point counts of `K_0` at `n = 3, 5, 9, 11, 23` equal the Koblitz recurrence (4, 44, 508, 2,116, 8,383,412).
- `x(P + T) = 1/x`: 0 mismatches for `P` and `-P`, exhaustive for `n <= 23` (4,191,705 points at `n = 23`), about 10^5 samples at each of `n = 29..41`.
- Exactness theorem (roots of `S_4(., ., ., x_R)` over `V^3` are rational decompositions unless two abscissae are non-liftable with `x_a = x_b, x_c = x_R` or `x_a x_b = x_c x_R = 1`): exhaustive for `V = F` at `n = 3, 5, 9` and for `wt <= 2, 3` at `n = 11` (1,057 targets each); 0 decompositions that are not roots; 0 unexplained extra roots; 0 extra roots at targets with `x_R` and `1/x_R` outside `V` (extra roots F1/F2: 6/3, 210/105, 65,274/32,637 for `V = F`; 968/0 and 13,068/2,178 for `wt <= 2, 3`).

3. Missing, partial or not done

- The panel's own oracle was not run for `(39, 3)` and `(41, 3)` (resource limit, not an impossibility: its pair-table code needs about 5 x 10^7 additions on the library's inversion, 449 s for the table at `(33, 3)` and 359 s at `(35, 3)`, and a full scan of the same size for each refuting target, 2,167 and 2,214 s at `(35, 3)`); at those two settings (16 targets) the verdicts and solution sets are checked against the example's own pair-table oracle, which reproduces the panel's `x_R`, verdict and `ref_count` on all 96 panel targets.
- Torsion at the three largest `w = 3` bases ran on 2 of 8 targets (a uniform and a planted one).
- Not attempted: a hardware-assisted normal-basis product; `T` on the repository's invariant subspace bases; the combined orbit-and-`T` loop; any `m = 2` or `m > 3` loop; any end-to-end cost.

4. Mechanism

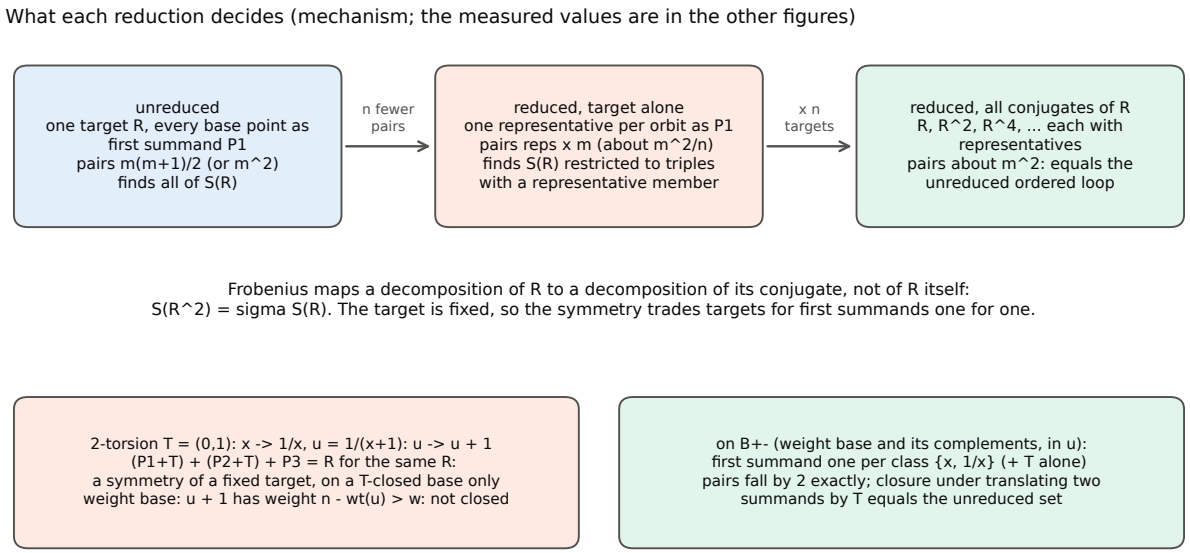


Figure 4: What each reduction decides. Frobenius maps decompositions of R to decompositions of its conjugates; translation by T maps decompositions of R to decompositions of the same R. Mechanism, not a measurement.